

WARSZAWA OKECIE, Poland

WMO: 123750

Lat: 52.1628N

Lon: 20.9611E

Elev: 106

StdP: 100.05

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.5	-12.2	-18.2	0.8	-14.8	-15.3	1.0	-11.4	12.3	5.0	10.9	2.8	2.9	90	0.558

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.9	30.2	20.3	28.4	19.4	26.8	18.7	21.6	27.9	20.7	26.3	19.7	24.8	3.6	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.5	14.5	23.8	18.7	13.7	23.0	17.9	13.0	22.3	63.2	28.1	60.0	26.3	56.9	24.9	25.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
9.6	8.4	7.5	DB	-17.9	33.4	3.6	1.7	-20.5	34.6	-22.6	35.6	-24.6	36.6	-27.3	37.8	(4)
			WB	-18.2	23.2	3.5	0.9	-20.8	23.8	-22.8	24.3	-24.8	24.8	-27.4	25.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.0	-1.9	-0.5	3.2	9.2	14.2	17.5	19.7	19.1	14.1	8.8	3.9	-0.3
	DBStd	8.72	5.53	5.04	4.43	4.32	3.96	3.49	3.29	3.19	3.45	4.07	4.23	5.10
	HDD10.0	1531	369	294	214	65	8	0	0	0	5	71	184	319
	HDD18.3	3586	628	528	471	275	138	57	23	29	133	294	432	577
	CDD10.0	1154	0	0	2	41	139	224	302	281	127	36	3	0
	CDD18.3	165	0	0	0	1	10	32	66	51	6	0	0	0
	CDH23.3	1457	0	0	0	18	113	291	562	422	52	0	0	0
	CDH26.7	390	0	0	0	2	18	72	172	117	9	0	0	0
Wind	WSAvg	3.6	4.1	4.0	4.1	3.7	3.5	3.5	3.2	3.0	3.2	3.5	3.7	3.8
Precipitation	PrecAvg	535	27	25	28	35	57	67	79	62	44	39	39	33
	PrecMax	798	79	72	65	93	183	149	295	179	138	148	109	82
	PrecMin	341	1	0	5	4	16	14	15	4	7	3	0	3
	PrecStd	96	15	16	13	19	28	34	50	34	27	31	20	19
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.1	11.6	17.4	25.5	28.9	31.5	33.2	32.8	27.9	21.9	14.9	11.1
		MCWB	7.7	8.2	10.5	15.0	18.1	20.9	20.9	21.0	18.6	16.6	11.8	9.4
	2%	DB	8.1	9.5	14.5	22.2	26.4	28.8	30.6	29.6	24.7	19.2	12.8	9.1
		MCWB	6.5	7.1	8.9	13.7	17.3	19.4	20.5	20.5	17.3	14.7	10.6	7.6
	5%	DB	6.4	7.8	12.1	19.4	24.1	26.6	28.5	27.5	22.5	17.0	11.1	7.7
		MCWB	5.0	6.0	7.9	12.1	16.2	18.3	19.8	19.5	16.4	13.5	9.3	6.5
	10%	DB	4.9	6.0	9.8	16.9	21.9	24.7	26.5	25.6	20.4	15.0	9.8	6.0
		MCWB	3.8	4.5	6.8	10.8	15.0	17.6	18.8	18.4	15.5	12.6	8.4	5.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.3	8.7	11.4	15.8	19.6	21.9	22.7	22.6	19.9	17.1	12.5	9.9
		MCDB	9.4	10.3	16.0	23.3	26.3	28.8	29.7	29.2	24.8	20.4	14.3	10.8
	2%	WB	6.5	7.5	9.7	14.4	18.2	20.6	21.7	21.4	18.3	15.3	11.0	7.8
		MCDB	7.7	9.1	13.5	21.2	24.4	26.6	28.2	27.6	23.1	18.1	12.4	8.6
	5%	WB	5.2	6.0	8.3	12.8	17.0	19.4	20.8	20.3	17.2	13.9	9.6	6.4
		MCDB	6.3	7.6	11.2	18.0	22.6	24.9	26.5	25.9	21.1	16.5	10.8	7.4
	10%	WB	3.7	4.5	7.0	11.5	15.8	18.4	19.9	19.4	16.0	12.7	8.4	5.0
		MCDB	4.7	5.9	9.5	15.7	20.7	23.2	24.8	24.2	19.6	14.9	9.5	5.9
Mean Daily Temperature Range	5% DB	MDBR	4.8	5.7	7.6	9.9	10.3	9.9	9.9	10.1	9.1	7.7	5.0	4.6
		MCDBR	5.6	7.7	11.6	14.1	13.5	13.0	12.9	13.0	12.3	10.7	7.2	5.6
		MCWBR	4.6	5.8	7.1	7.0	6.2	5.7	5.0	5.6	6.0	6.6	5.3	4.6
	5% WB	MCDBR	5.4	7.0	10.5	12.5	12.1	11.6	11.2	11.7	10.9	9.8	6.5	5.3
		MCWBR	4.6	5.6	7.1	6.6	6.3	5.9	5.1	5.4	5.9	6.5	5.2	4.6
Clear-Sky Solar Irradiance	taub	0.322	0.351	0.385	0.430	0.416	0.411	0.442	0.435	0.396	0.373	0.361	0.318	
	taud	2.225	2.211	2.198	2.141	2.241	2.282	2.212	2.231	2.309	2.338	2.274	2.264	
	Ebn at Noon	650	743	794	802	837	843	807	790	777	714	598	589	
	Edh at Noon	69	93	115	137	130	126	133	124	102	82	67	57	
All-Sky Solar Radiation	RadAvg	0.69	1.42	2.63	4.09	5.18	5.49	5.12	4.49	3.18	1.77	0.74	0.50	
	RadStd	0.14	0.23	0.35	0.56	0.53	0.48	0.62	0.37	0.48	0.28	0.07	0.07	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		+0.69	N/A	N/A	N/A	N/A	N/A	-149	-229	+115	+37
Regional (24 neighbors)		+0.49	N/A	N/A	N/A	N/A	N/A	N/A	-165	+78	+28

Nomenclature: See separate page